Cutaneous Biology & Applied Pharmaceutical Sciences

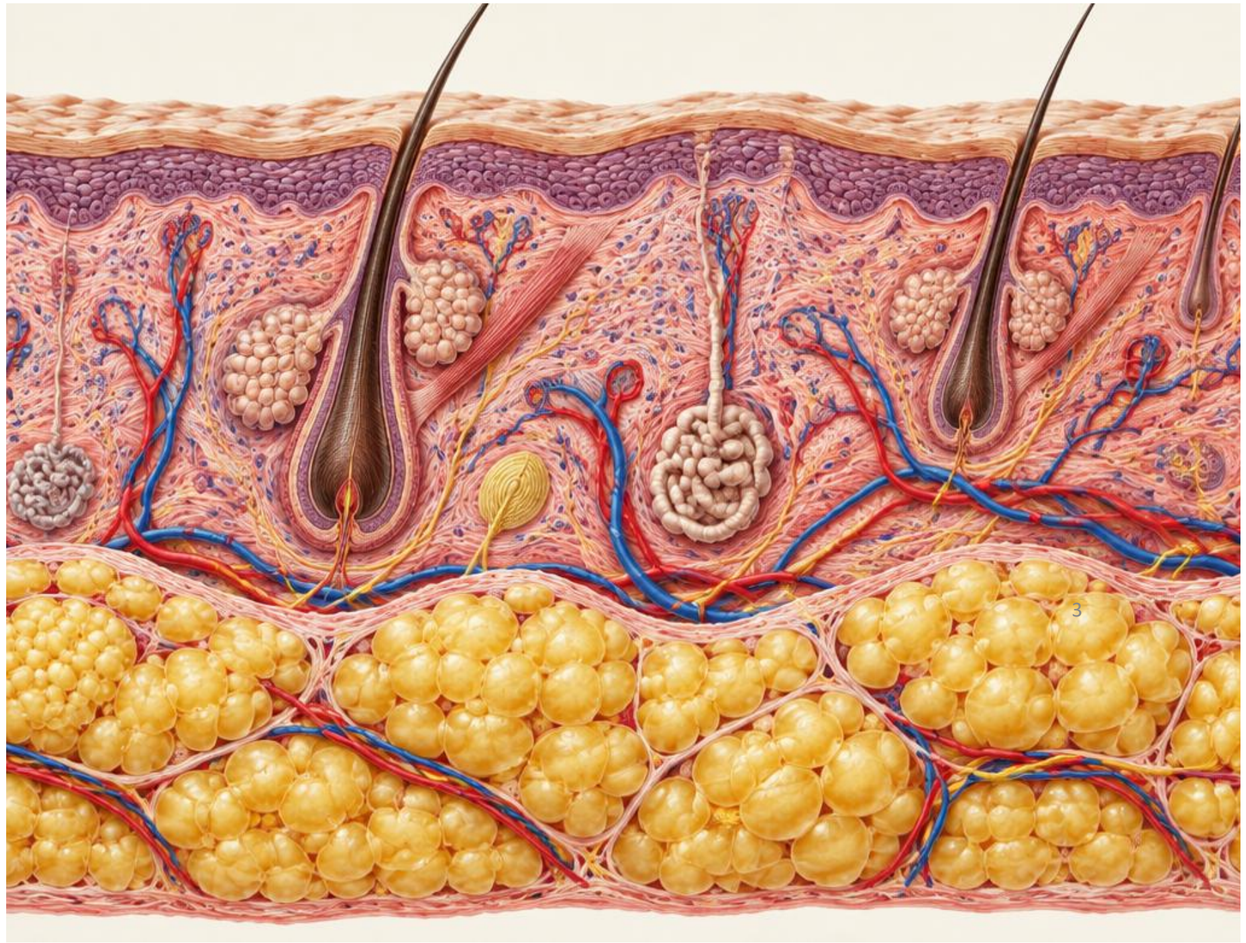


PART 1

Cutaneous Structure and Physiological Functions



Skin structure



EPIDERMIS

Avascular barrier, outermost defensive barrier.

DERMIS

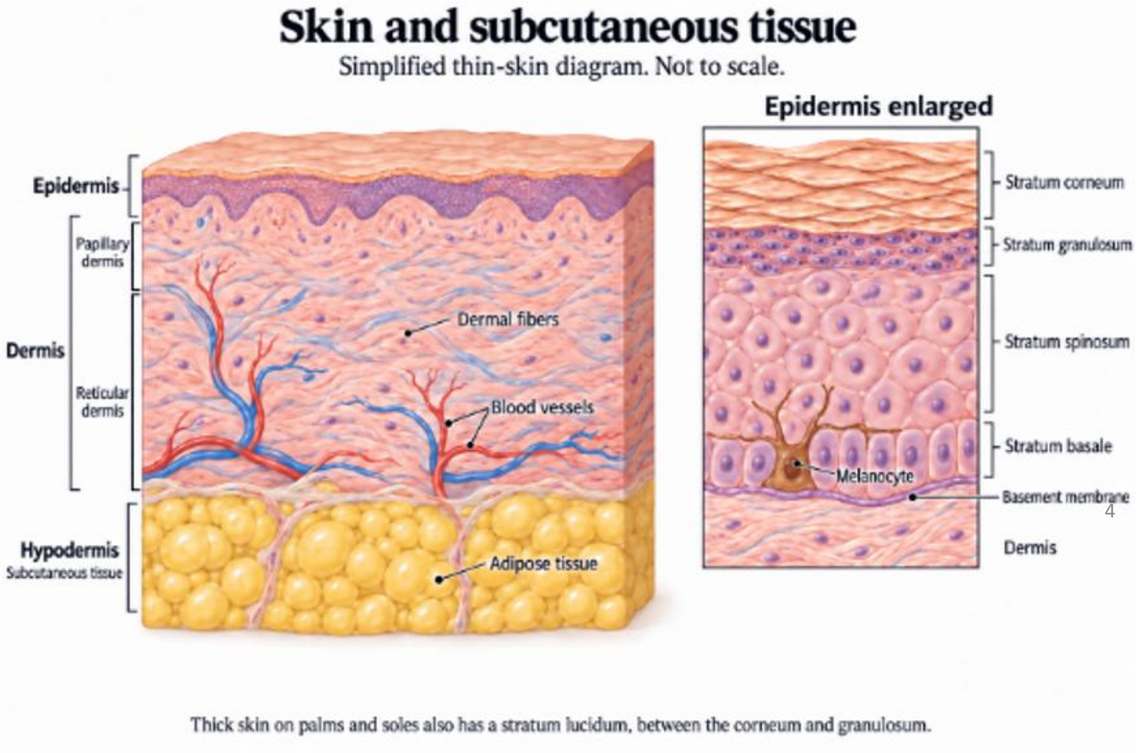
Collagen, elastin and a microvascular network; key local target and systemic uptake interface.

HYPODERMIS

Adipose depot; metabolic and pharmacological reservoir.

APPENDAGES

The Epidermis: *Layered Frontline Barrier*



STRATUM CORNEUM

"Brick + mortar": corneocytes within ceramide–cholesterol–fatty acid lamellae.

STRATUM GRANULOSUM

Lipid processing and tight junctions refine paracellular resistance.

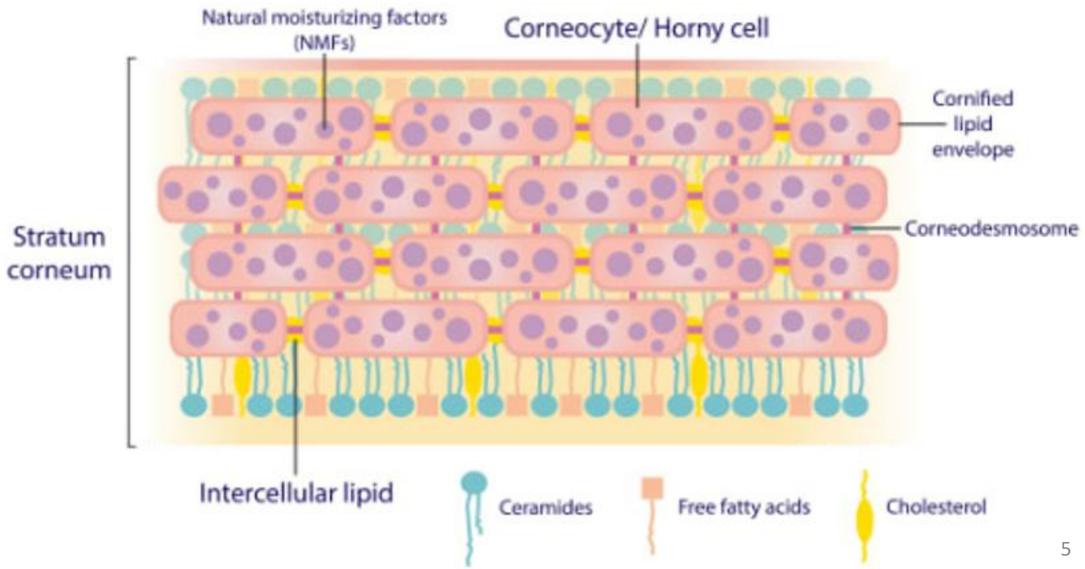
STRATUM SPINOSUM

Mechanical cohesion; immune surveillance.

STRATUM

Stem-cell renewal, melanocytes, mitosis; pigmentation target zone.

The Epidermis: *The Brick and Mortar Paradigm*



Structure of stratum corneum

CORNEOCYTES ("BRICKS")

Corneocytes

INTERCELLULAR LIPIDS ("MORTAR")

Equimolar lamellar matrix of ceramides, cholesterol, and free fatty acids

Acid Mantle Buffer (pH 4.5–5.5): Microbial inhibition

Supporting Microenvironment: *Dermis & Hypodermis*



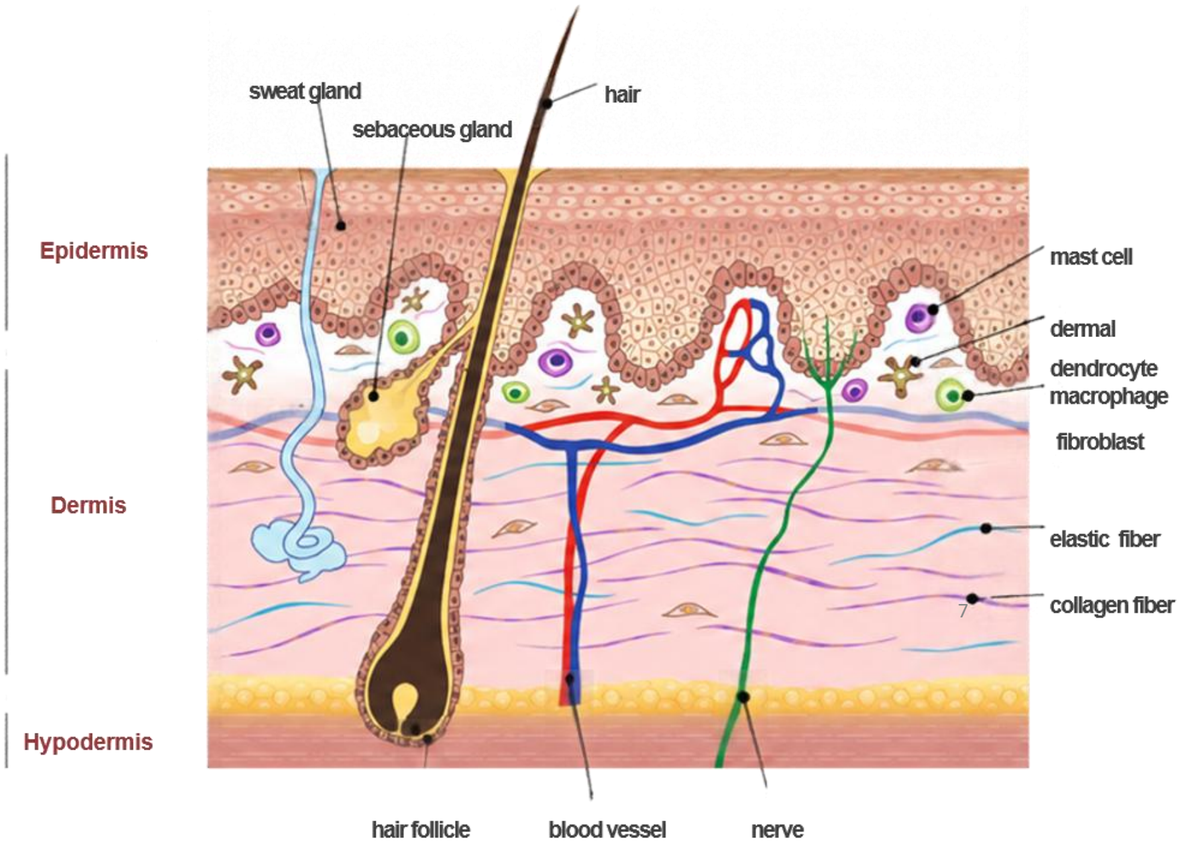
HYDROPHILIC DERMAL MATRIX

Hyaluronic acid retains water; extracellular matrix can slow lipophilic solutes yet accommodate polar species.

LIPOPHILIC DEPOT

Adipocytes : store high-log P drugs,

Dermis



Collagen: Tensile Strength

Elastin: Elasticity

Hyaluronic acid: Moisture

Dense vascular and neural network

Appendages

Hair Follicles

Physical protection and tactile transmission.

Sebaceous Glands

Secret natural lipids to soften skin surface.

Sweat Glands

Cool surface and regulate body temperature.

Nails

Protect fingertips and toes from injury.

Core Cutaneous Functions

PROTECTIVE BARRIER

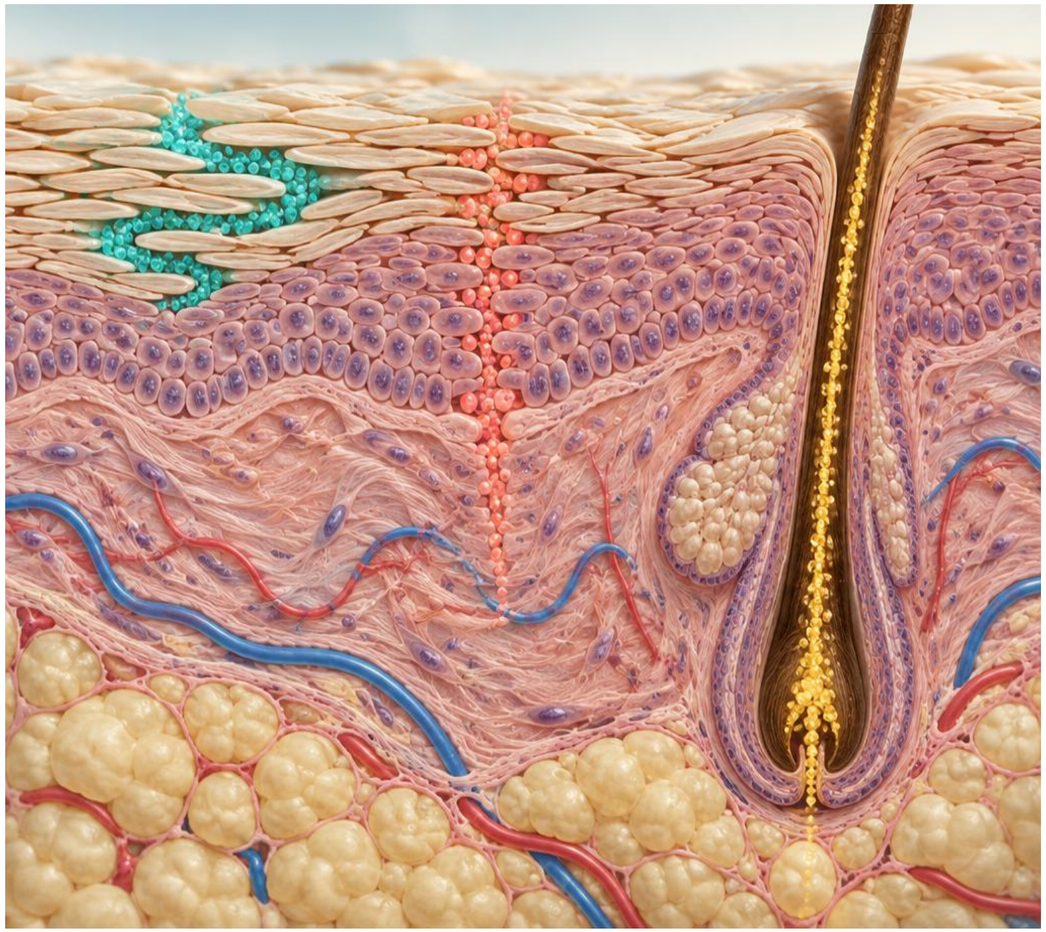
THERMOREGULATION

SENSORY TRANSDUCTION

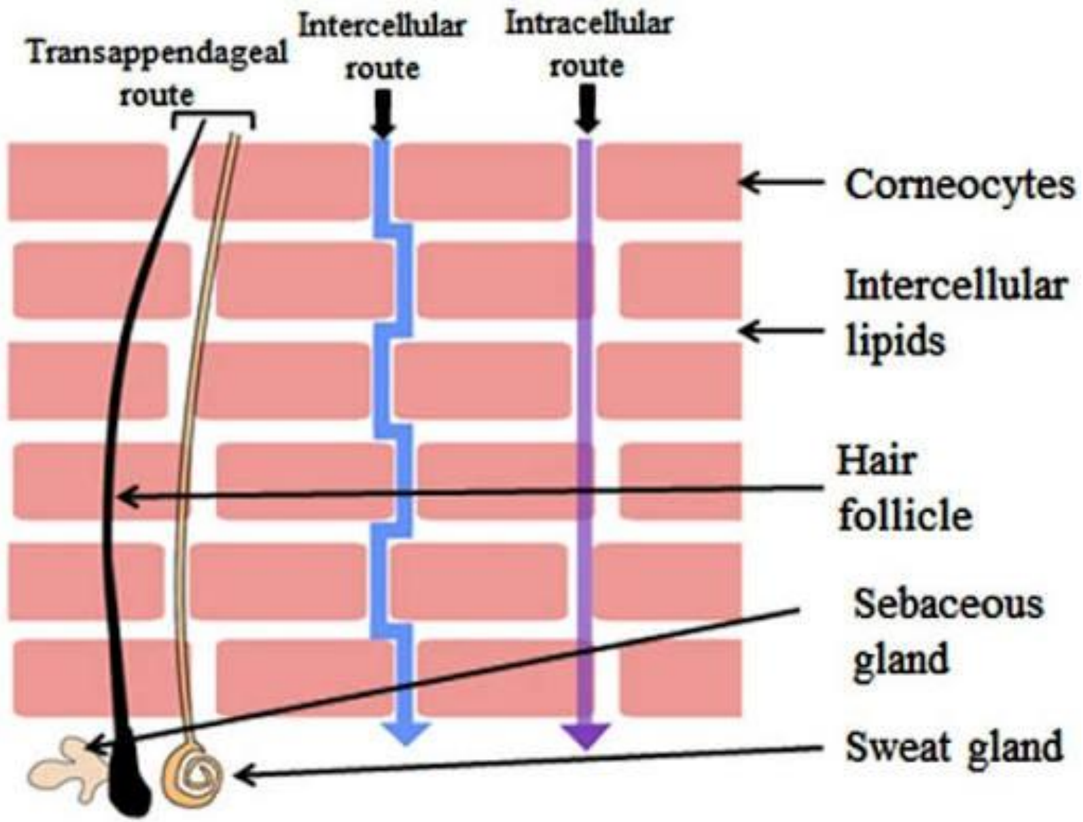
SYNTHESIS

PART 2

Percutaneous Transport Kinetics



Percutaneous Penetration Pathways



INTERCELLULAR

Tortuous lipid-matrix route; dominant for lipophilic substances.

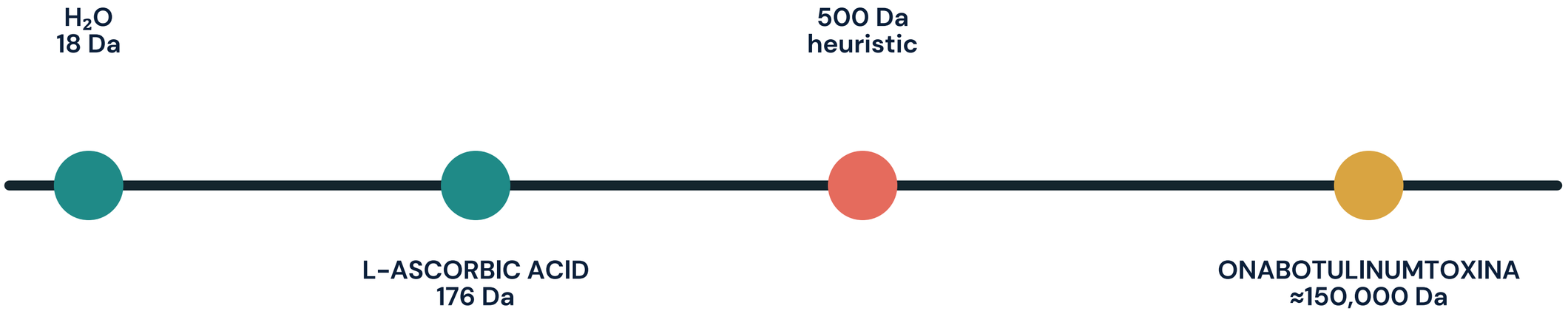
TRANSCELLULAR

Repeated polar/non-polar partitioning through corneocytes.

TRANSAPPENDAGEAL

Follicular and glandular.

The Bos–Carbatsis *500-Dalton* Paradigm



Octanol- water partition coefficient (logP) between 1 and 3

Biopharmaceutical Targeting in Formulation

SURFACE COVER

High-MW film formers at the interface.

SKIN CARE

Intercellular / transcellular/ Transappendageal

SYSTEMIC TRANSDERMAL

Polymeric patches + enhancers sustain input to dermal vessels.

Advanced Drug Delivery Platforms

Vehicle microstructure controls localization, release kinetics and tolerability.

EMULSIONS

O/W ↔ W/O

Phase continuity, occlusion and barrier feel

VESICLES

Liposomes / niosomes

Dual-domain encapsulation; membrane interaction

HYDROGELS

Serums / networks

Fast wetting and burst release

MICROSPHERES

Polymeric depots

Sustained release; irritation mitigation

Strategies to Bypass the 500 Dalton Rule

NANOCARRIERS

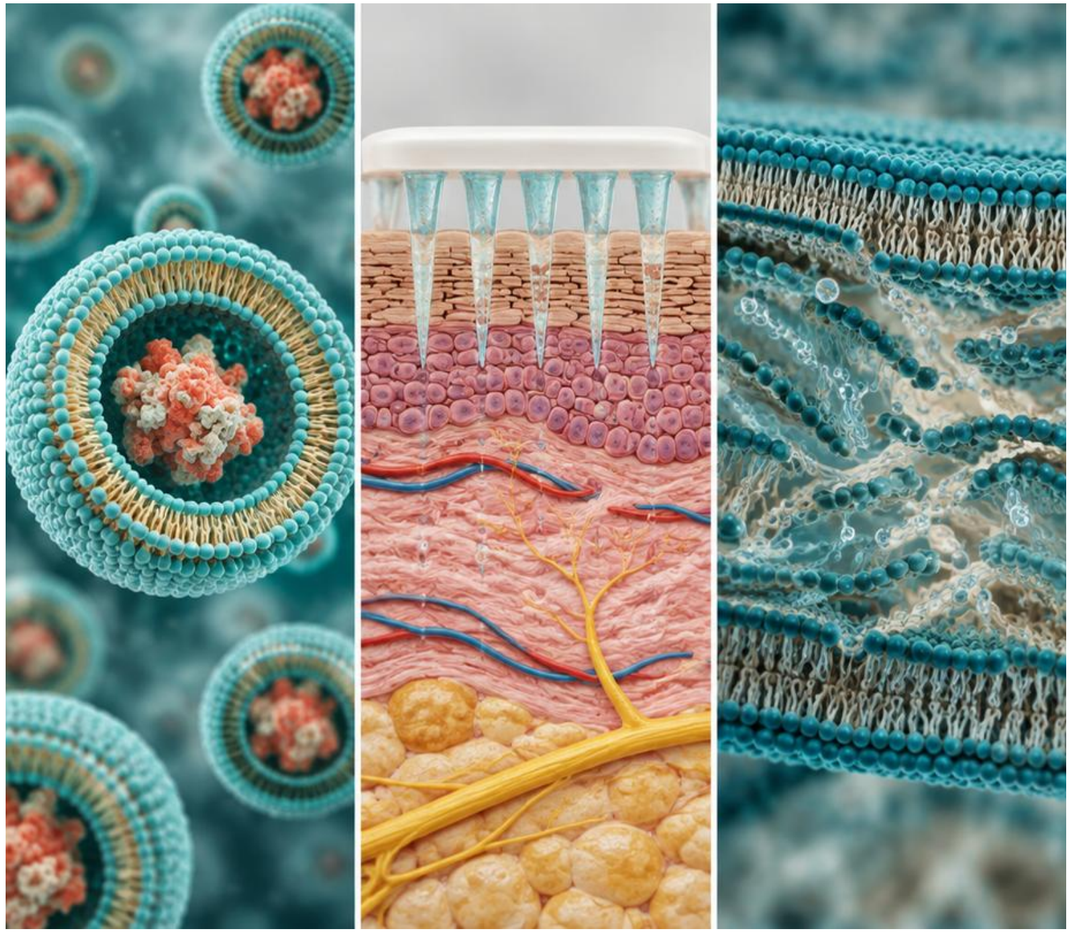
Liposomes protect and localize macromolecular payloads; apparent delivery depends on carrier fate.

MICRONEEDLE ARRAYS

Physical microchannels cross the barrier while remaining short of deep nociceptors.

CHEMICAL ENHANCERS

Ethanol and propylene glycol alter solubility and create reversible, transient pathways.



PART 3

Pathogenesis & Pharmacotherapy

Acne Vulgaris Pathophysiology

anaerobic environment

1 HYPERSEBORRHEA

Androgen sensitive sebocytes increase lipid output.

2 FOLLICULAR HYPERKERATINIZATION

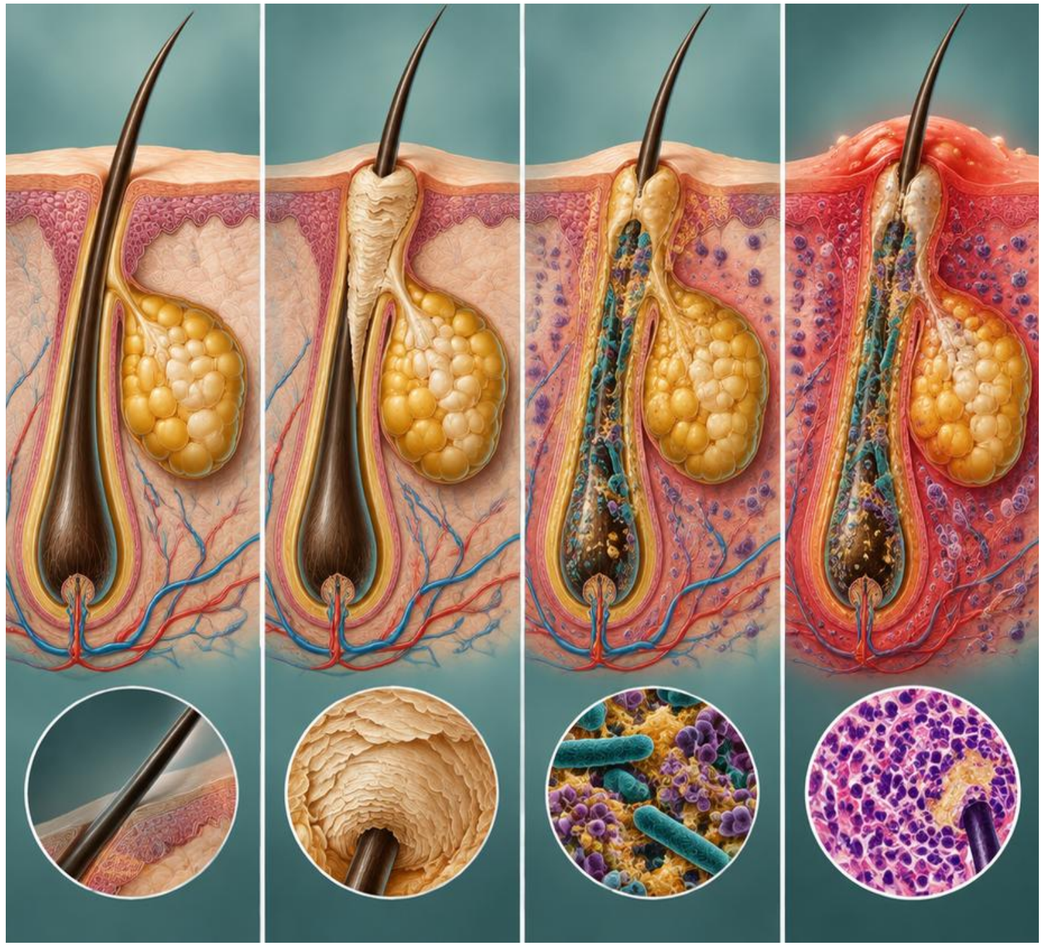
Corneocyte retention occludes follicles.

3 MICROBIAL GROWTH

Anaerobic follicle favors *Cutibacterium acnes*.

4 INFLAMMATORY ACTIVATION

Innate signaling recruits cytokines and neutrophils(→ Pustucle).



Acne Pharmacotherapy Strategies

SEBO-SUPPRESSION

Retinoids / isotretinoin

BACTERICIDAL

Benzoyl peroxide

EXFOLIATION

Salicylic acid (BHA)

ANTI-INFLAMMATORY

Azelaic acid

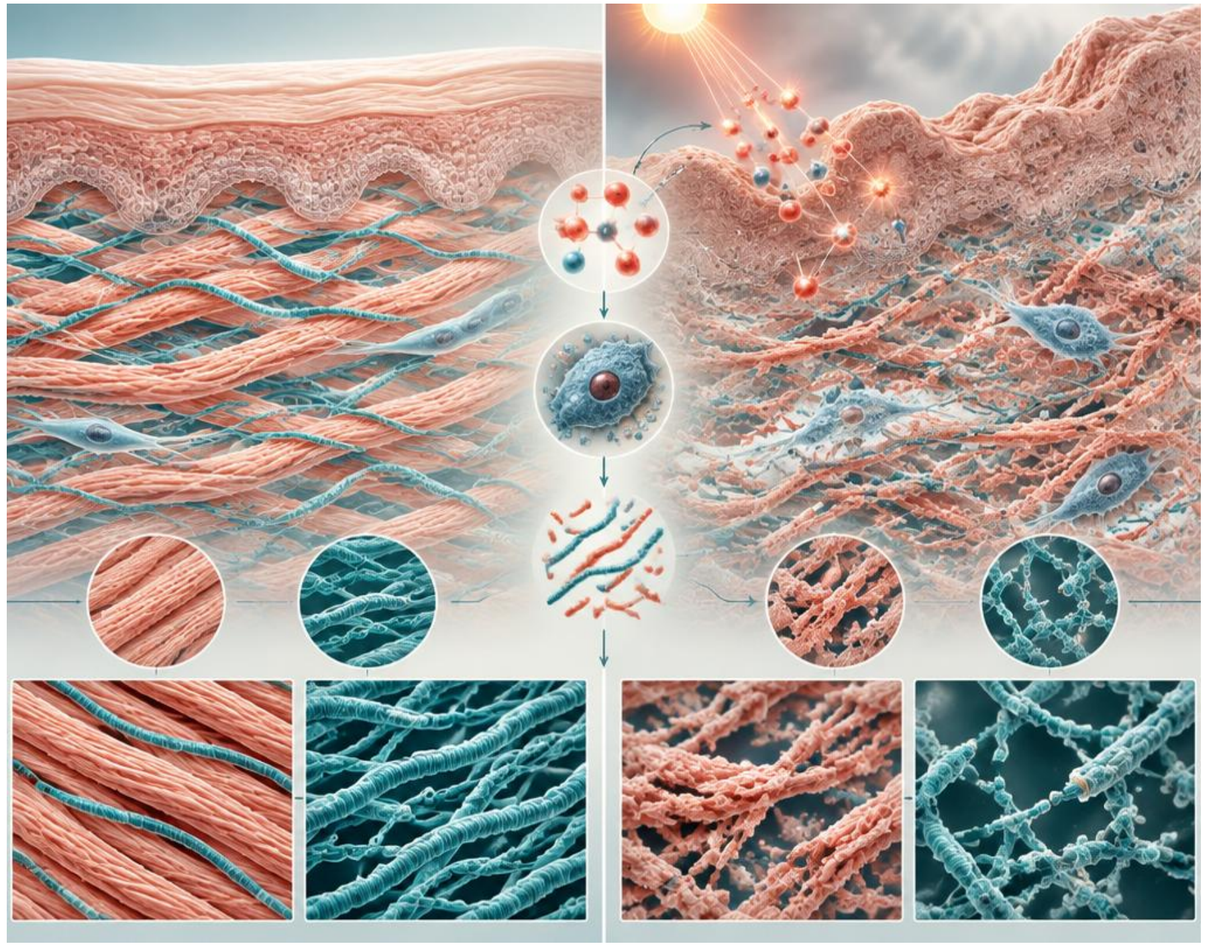
Mechanisms of Cutaneous Aging

INTRINSIC

Telomeric attrition → fibroblast senescence → reduced collagen and matrix repair.

EXTRINSIC / PHOTOAGING

UV → ROS → AP-1 signaling → MMP overexpression → collagen fragmentation.



Anti-Aging Pharmacotherapeutic Approaches

PHOTOPROTECTION

Physical + chemical UV filters; radical scavengers limit upstream ROS.

MOLECULAR SYNTHESIS

Retinoids and biomimetic peptides promote collagen transcription and remodeling.

BARRIER RECONSTITUTION

Bio-identical lipids restore lamellae; hyaluronic acid supports hydration.